Jan Kazimierczak

Engineering Science (aerospace focus), U of T · Canadian study permit: eligible for full-time summer work in Canada
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Languages: English (C1, fluent) · Polish (native) · Spanish (native)

Second-year Engineering Science student (intended Aerospace specialization) on the UTAT Rocketry hybrid engine team, with hands-on experience taking mechanical hardware from CAD (Fusion 360, Onshape) to 3D-printed parts, custom test rigs and physical testing. Seeking a mechanical engineering internship, May–August 2027.

Engineering Team Experience

University of Toronto Aerospace Team (UTAT) – Rocketry, Hybrid Engine / Propulsion Toronto, Canada
Team Member *Jan 2026 – Present*

- Assembled the hybrid engine cold-flow test setup and connected its instrumentation, working alongside the test lead.
- Helped improvise a fix for a test-stand issue that came up during cold-flow testing.
- Assigned to machine the hybrid engine injector once its design is finalized.
- Authored a report comparing four paraffin/N_2O hybrid fuel compositions using NASA CEA and literature; recommended keeping the current fuel as a reference, prioritizing carbon-black evaluation and deferring aluminium additives.
- Ongoing: ANSYS Fluent CFD analysis of hybrid engine nozzle designs.
- Earlier: background research on regenerative cooling for the team's liquid rocket engine.

Mechanical Design, Build & Test Projects

Grid Fin Aerodynamics and Wind Tunnel Testing · *European Schools Science Symposium* *Sep 2023 – Apr 2024*

- Designed 18 grid-fin variants in Fusion 360 and 3D printed them for controlled testing of hole geometries.
- Designed, built and assembled a custom 3 m wind tunnel (honeycomb straightener, contraction, 30 cm test section, diffuser, fan).
- Built and instrumented an electro-mechanical load-cell test rig (TAL-220 load cell, HX711, Arduino Uno); captured 180,000+ sensor readings through a Python-to-CSV acquisition pipeline.
- Analyzed test data in R (ANOVA, Tukey HSD) and co-authored the three-person team report, documenting the result's model-scale limits; the simple-outline 6 mm square-hole fin produced the highest measured drag force, which is consistent with SpaceX's Falcon 9 grid fin geometries.

Matboard Bridge Structural Optimization · *CIV102, U of T, 3-person team* *Nov 2025*

- Compared bridge geometries with Python simulations of a moving train load, verifying selected outputs with three hand calculations before committing to fabrication.
- Raised the model-predicted minimum failure load from 240 N to 870 N (3.6×) over four design iterations.
- Fabricated and load-tested the 1.265 m bridge; post-test analysis traced failure to a splice/glue joint absent from the analytical model.

Engineering Design Projects · *Praxis I & II, U of T* *Sep 2025 – Apr 2026*

- BikePack Buddy (Praxis I): designed and manufactured a full-scale prototype. Designed tests to verify the prototype met the requirements.
- Screw sorter (Praxis II): trained and tested a CNN for screw classification and integrated camera-based inference on a Raspberry Pi. Designed and built the test rig and generated synthetic images for training.

Education

University of Toronto Toronto, Canada
BASc, Engineering Science (intended Aerospace specialization), 2nd year, incl. PEY co-op *Expected May 2030*

- Cumulative GPA: **3.58/4.00**.
- Relevant coursework (Fall 2026 / Winter 2027): Thermodynamics & Heat Transfer; Ordinary Differential Equations; Praxis III (mechatronic design); Probability & Statistics; Vector Calculus & Fluid Mechanics.

Technical Skills

CAD & drawings: Fusion 360, Onshape (part design, engineering drawings)
Build & test: fabrication (including machining) and assembly, 3D printing, test-rig building, load testing, sensor instrumentation & data acquisition (Arduino, HX711, Raspberry Pi)
Analysis: structural hand calculations, ANSYS Fluent CFD, NASA CEA
Programming: Python, MATLAB, C, System Verilog, RISC-V assembly

Additional Experience

Leadership & community: IT Officer, Pupils' Committee, European School of Alicante (2024–25); community kitchen volunteer, Alicante Gastronómica Solidaria (2024); youth basketball coach, Fundación Lucentum Alicante (2023).